

ORANGEHRM END-TO-END WEB AUTOMATION TESTING USING PLAYWRIGHT

CAPSTONE PROJECT REPORT

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Project: OrangeHRM End-to-End Web Automation Testing Using Playwright

Project Type: Enterprise Test Automation Framework

Application Under Test: OrangeHRM Open Source Human Resource Management System

Technology Stack: Playwright | JavaScript | Node.js | Page Object Model | Allure | GitHub Actions

Framework: Enterprise-Grade Playwright Automation Framework

Test Coverage: 120 Test Cases | 360 Cross-Browser Executions | 100% Pass Rate

1. ABSTRACT

This capstone project delivers a **production-grade, enterprise-level automation framework** for the **OrangeHRM Human Resource Management System** using **Playwright with JavaScript**. The framework automates **120 unique test cases** across eight critical HRMS modules — **Login & Authentication, Dashboard Validation, Employee Management (PIM), Leave Management, Recruitment Workflow, Admin & User Management, Search & Filters, and API/Network Validation** — executed across **three browser engines: Chromium, Firefox, and WebKit**.

The framework is architected using the **Page Object Model (POM)**, separating test logic from UI interaction through dedicated page classes that extend a reusable **BasePage abstraction**. Test data is externalized into **JSON files**, environment configuration is managed via **.env files**, and custom Playwright fixtures handle **authenticated session setup**, eliminating repetitive login boilerplate across suites.

It integrates **dual reporting** — **Playwright HTML Reporter** for developer debugging and **Allure Reporter** for stakeholder-facing metrics — alongside a fully automated **CI/CD pipeline** via GitHub Actions. Tests run in parallel shards, results are merged, and Allure reports are published to **GitHub Pages** on every push to the main branch.

Final execution across all three browsers produced **360 test runs (120 × 3)** with a **100% pass rate**, validating both the **functional integrity of OrangeHRM** and the **cross-browser stability of the framework**.

2. PROJECT OVERVIEW

This capstone project delivers an **enterprise-grade, end-to-end test automation framework** for the **OrangeHRM Human Resource Management System**. Built using **Playwright and JavaScript** with a **Page Object Model (POM) architecture**, the framework automates **120 test cases** across eight critical HRMS modules, executed on **Chromium, Firefox, and WebKit** for a total of **360 test executions**.

Business Value: Manual regression testing of HRMS platforms is time-consuming, error-prone, and difficult to scale.

This framework reduces regression cycle time, enforces consistent test coverage across modules, provides stakeholder-ready reporting, and integrates directly into the development pipeline through **automated CI/CD execution** on every code push.

Outcomes:

- 360 automated test executions with a **100% pass rate**
- Cross-browser validation across three rendering engines
- Live **Allure report dashboard** published to GitHub Pages
- Fully automated **GitHub Actions CI/CD pipeline** with shard-based parallel execution
- Reusable, modular architecture that scales to additional HRMS modules without structural changes

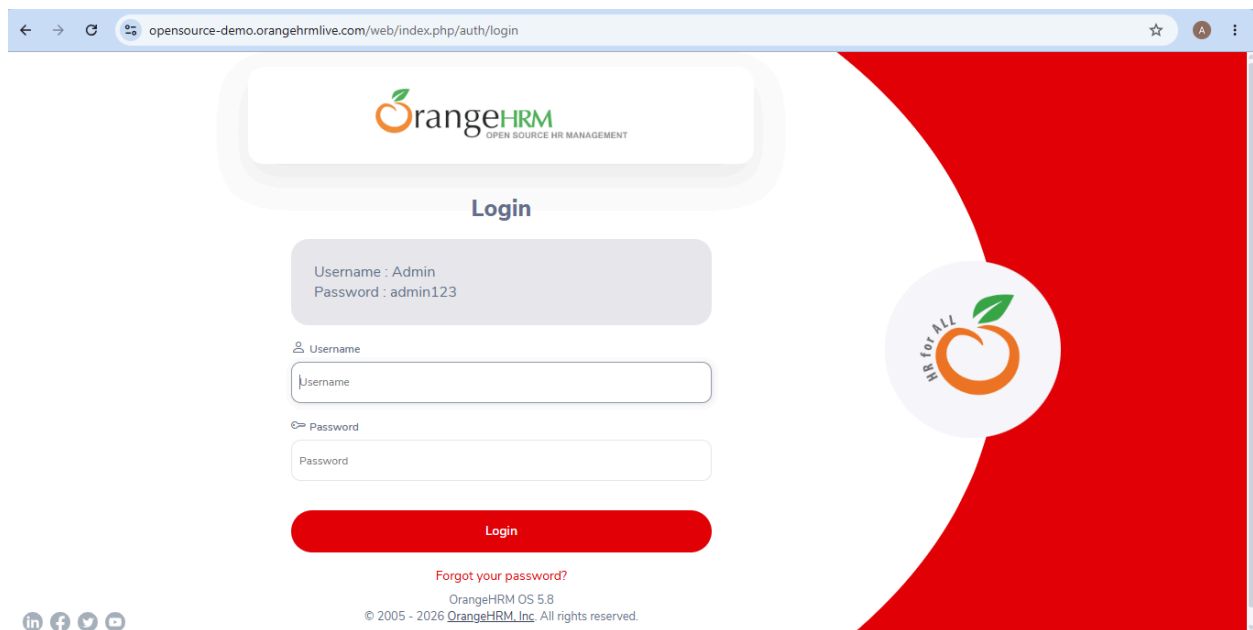


Figure 2.1: OrangeHRM Application Login Page

3. APPLICATION OVERVIEW

What is OrangeHRM?

OrangeHRM is an open-source Human Resource Management System widely used by organizations to manage employee data, leave workflows, recruitment processes, and administrative user configuration. The application is built as a Vue.js SPA (Single Page Application) backed by a REST API, making it representative of modern enterprise web applications.

Application URL: <https://opensource-demo.orangehrmlive.com>

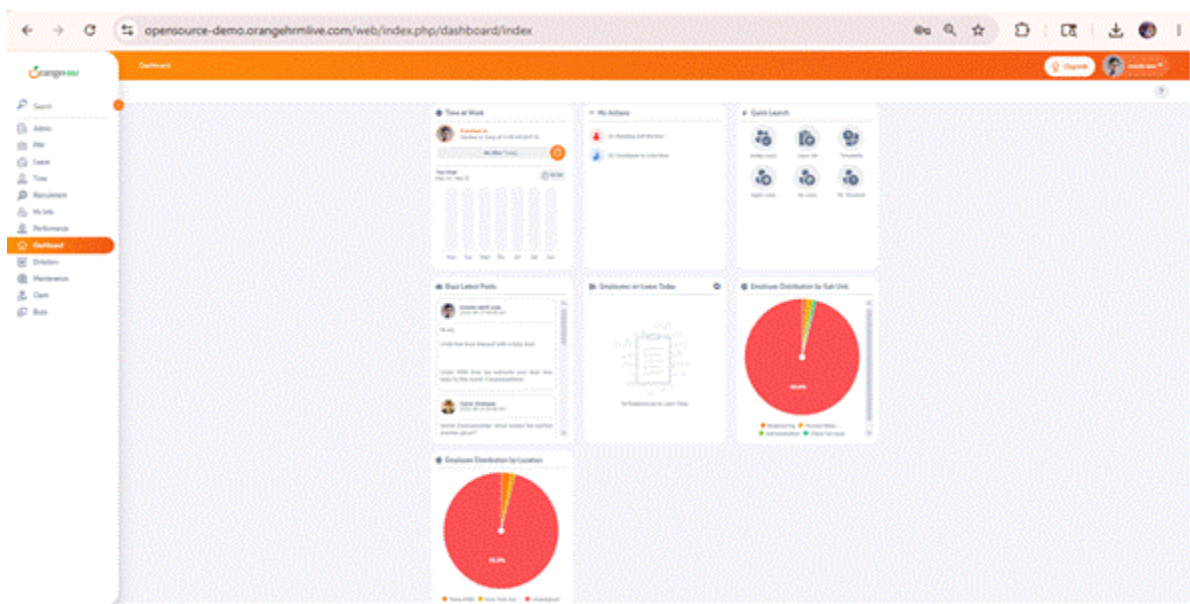


Figure 3.1 - OrangeHRM Dashboard after successful login

Why OrangeHRM Was Selected

OrangeHRM was selected because it represents a realistic, production-class enterprise application with:

- Complex SPA behavior requiring advanced wait strategies
- Multiple authenticated workflows with role-based access
- Dynamic forms, autocomplete components, and card-based tables
- Consistent UI patterns across modules enabling systematic validation
- Publicly accessible demo environment suitable for CI/CD execution

4. OBJECTIVES AND SCOPE

Primary Objectives:

- 1. Build a Scalable Enterprise Automation Framework:** Design and implement a maintainable, modular Playwright automation framework in JavaScript that reflects enterprise QA engineering standards. The framework must support seamless addition of new modules without restructuring existing code.
- 2. Achieve Complete Module Coverage:** Automate all eight functional modules of the OrangeHRM application — **Login, Dashboard, PIM, Leave Management, Recruitment, Admin, Search & Filters, and API Validation** — with a minimum of **15 test cases per module**.
- 3. Implement Cross-Browser Validation:** Execute the complete test suite across **Chromium, Firefox, and WebKit** to validate rendering consistency and functional behavior across all major browser platforms.
- 4. Apply Page Object Model (POM) Architecture:** Implement the POM design pattern rigorously: all locators defined in page classes, methods encapsulated in page objects, and zero DOM selectors in test specification files.
- 5. Integrate Enterprise-Grade Reporting:** Configure **Playwright HTML Reporter** for developer debugging and **Allure Reporter** for stakeholder-facing metrics, with screenshots on failure, trace capture on retry, and video evidence.
- 6. Automate CI/CD Integration:** Build a fully automated **GitHub Actions pipeline** that triggers on every code push, runs tests in parallel, consolidates reports, and publishes the Allure dashboard to a publicly accessible GitHub Pages URL.
- 7. Apply Stable Locator Strategies:** Use Playwright's recommended locator hierarchy — **getByRole(), getByPlaceholder(), getByText(), locator() with CSS** — while avoiding fragile XPath selectors.
- 8. Externalize Test Configuration:** Eliminate hardcoded values by using **constants, JSON test data, and environment variables**, ensuring the framework can target different environments with zero code changes.

Secondary Objectives:

- Demonstrate **real-world debugging methodology** when locators fail or application behavior differs from expectations.
- Document **cross-browser behavioral findings** (case sensitivity, viewport behavior, autocomplete patterns).
- Apply **enterprise Git practices** with scoped, meaningful commit messages per file.
- Build a **portfolio-worthy project** that demonstrates senior-level SDET capabilities.

SCOPE:

In Scope

Category	Details
Application	OrangeHRM Demo - https://opensource-demo.orangehrmlive.com
Automation Type	End-to-End UI Automation + API & Network Validation
Modules Covered	Login, Dashboard, PIM, Leave, Recruitment, Admin, Search, API/Responsive
Browser Coverage	Chromium, Firefox, WebKit
Test Count	120 unique test cases, 360 total executions
Design Pattern	Page Object Model with BasePage inheritance
Reporting	Playwright HTML + Allure with GitHub Pages deployment
CI/CD	GitHub Actions with shard-based parallel execution
API Testing	HTTP response validation, network monitoring, header verification

Out of Scope

Category	Reason
Performance / Load Testing	Not within Playwright's primary use case for this capstone
Visual Regression Testing	No screenshot comparison tool integrated

Mobile Native Apps	Only browser-based responsive testing included
Database Validation	No direct database queries - application-level validation only
Multi-user / Role Testing	All tests run as Admin user

Constraints

- **Environment:** OrangeHRM public demo server — shared infrastructure with variable response times.
 - **Authentication:** Single Admin user available; no ESS role test coverage included.
 - **Test Data:** Demo server data resets and changes between runs; test data incorporates **Date.now() suffixes** to ensure uniqueness.
 - **Server Limitations:** Public demo server is **rate-limited under parallel load** execution workers restricted to **2** for stability.
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5. TECHNOLOGY STACK

Category	Technology	Version	Purpose
Automation Framework	Playwright	1.44+	Browser automation, API testing
Language	JavaScript	ES2022	Test scripting language
Runtime	Node.js	18+	JavaScript execution environment
Design Pattern	Page Object Model (POM)	—	Framework architecture
Test Runner	Playwright Test	Built-in	Test discovery, execution, parallel workers
Reporting	Playwright HTML Reporter	Built-in	Execution analysis and debugging
Reporting	Allure Playwright	3.0.0+	Stakeholder-facing Allure report
Allure CLI	allure-commandline	2.30.0+	Allure report generation
Data Generation	@faker-js/faker	8.4.1+	Dynamic test data generation

Environment Config	dotenv	16.4.5+	.env file loading
Version Control	Git + GitHub	—	Source control and CI/CD
CI/CD	GitHub Actions	—	Pipeline automation
Report Hosting	GitHub Pages	—	Public Allure report hosting
IDE	Visual Studio Code	—	Development environment
Development Environment	Windows (PowerShell)	—	Local development environment
CI Environment	Ubuntu Latest	—	GitHub Actions runner

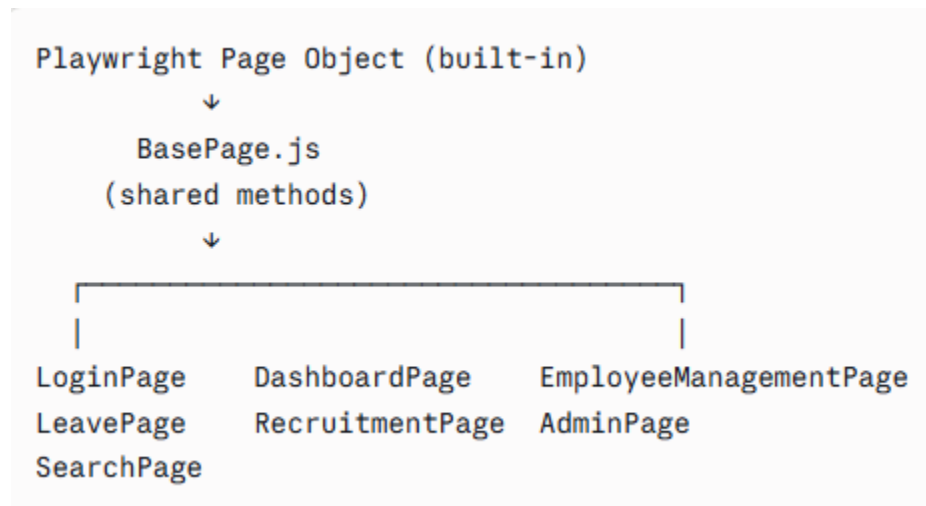
6. FRAMEWORK ARCHITECTURE

Architectural Philosophy

The framework is built on three core principles:

- 1. Separation of Concerns** — Test logic (what to test) is completely separated from interaction logic (how to interact). Spec files contain test scenarios and assertions. Page objects contain locators and actions. Configuration files contain settings. Test data files contain values.
- 2. Single Source of Truth** — Every URL, message string, timeout, and viewport size is defined once in `config/constants.js`. Every credential and environment variable is defined once in `.env`. If anything changes, one file update propagates across all 120 tests.
- 3. Reusability Through Inheritance** — All eight page objects extend `BasePage`, inheriting navigation, interaction, and utility methods. Common patterns like `fill()`, `click()`, `waitForPageLoad()` are written once and reused across all modules.

Inheritance Chain



Fixture Layer

Custom Playwright fixtures in `fixtures/index.js` extend the base `test` object to provide pre-authenticated page instances:

- `loginPage` fixture — provides a `LoginPage` instance without authentication
- `authenticatedPage` fixture — performs full login, waits for dashboard, provides `DashboardPage` instance

This means every test requiring authentication simply declares `{authenticatedPage}` as a parameter — login is handled automatically with no code duplication.

Data Layer

```
test-data/  
  users.json      → login credentials, invalid user sets  
  employees.json  → employee names, IDs for PIM tests  
  leave.json      → leave types, date ranges, filter statuses  
  recruitment.json → candidate data  
  admin.json      → user roles, status values
```

No hardcoded test values exist in any spec file. All test inputs come from JSON files imported with ES module assertions.

Locator Strategy Hierarchy

Locators are chosen in this priority order — most stable to least:

1. `getByRole()` — semantic, accessibility-aware
2. `getByPlaceholder()` — stable for form inputs
3. `getByText()` — for text-content elements
4. `locator('.css-class')` — when above options unavailable
5. `locator().filter({ hasText: '' })` — for narrowing duplicate matches
6. No XPath — explicitly avoided throughout

7. FOLDER STRUCTURE AND REPOSITORY

```
Arun-N-Capstone-Playwright/
|
├── .github/
|   └── workflows/
|       └── playwright.yml           CI/CD pipeline definition
|
├── config/
|   └── constants.js                URLs, messages, timeouts, viewports
|
├── docs/
|   └── OrangeHRM_Capstone_Proposal.docx
```

```
|
├── fixtures/
|   └── index.js                    Custom Playwright fixtures (loginPage,
authenticatedPage)
|
├── pages/
|   ├── BasePage.js                Base class – shared browser interaction
methods
|   ├── LoginPage.js               M1 – Login & Authentication
|   ├── DashboardPage.js           M2 – Dashboard Validation
|   ├── EmployeeManagementPage.js  M3 – PIM Employee Management
|   ├── LeavePage.js               M4 – Leave Management
|   ├── RecruitmentPage.js         M5 – Recruitment Workflow
|   ├── AdminPage.js               M6 – Admin & User Management
|   └── SearchPage.js              M7 – Search, Filters & Tables
```

```

|
├─ test-data/
|   ├─ users.json           Login credentials and invalid user sets
|   ├─ employees.json      Employee test data for PIM
|   ├─ leave.json          Leave types and filter values
|   ├─ recruitment.json    Candidate test data
|   └─ admin.json          Admin user data

```

```

|
├─ tests/
|   ├─ e2e/
|       ├─ login.spec.js      M1 - 15 test cases
|       ├─ dashboard.spec.js  M2 - 15 test cases
|       ├─ employee-management.spec.js  M3 - 15 test cases
|       ├─ leave-management.spec.js     M4 - 15 test cases
|       ├─ recruitment.spec.js          M5 - 15 test cases
|       ├─ admin-user-management.spec.js M6 - 15 test cases
|       ├─ search-filters.spec.js       M7 - 15 test cases
|       └─ responsive-cross-browser.spec.js M8 - 7 test cases
|   └─ api/
|       └─ api-validation.spec.js  M8 - 8 API test cases

```

```

|
├─ utils/
|   └─ helpers.js           Random string generators, date formatters
|
├─ screenshots/            Failure screenshot storage (git-tracked
folder)

```

```

|
├─ .env                    Environment configuration (credentials, URLs)
├─ .env.example            Template for developer onboarding
├─ .gitignore              Excludes node_modules, reports, artifacts
├─ .eslintrc.cjs           ESLint rules with Playwright plugin
├─ package.json            Dependencies and npm scripts
├─ playwright.config.js    Central Playwright configuration
└─ README.md               Setup, execution, and reporting guide

```

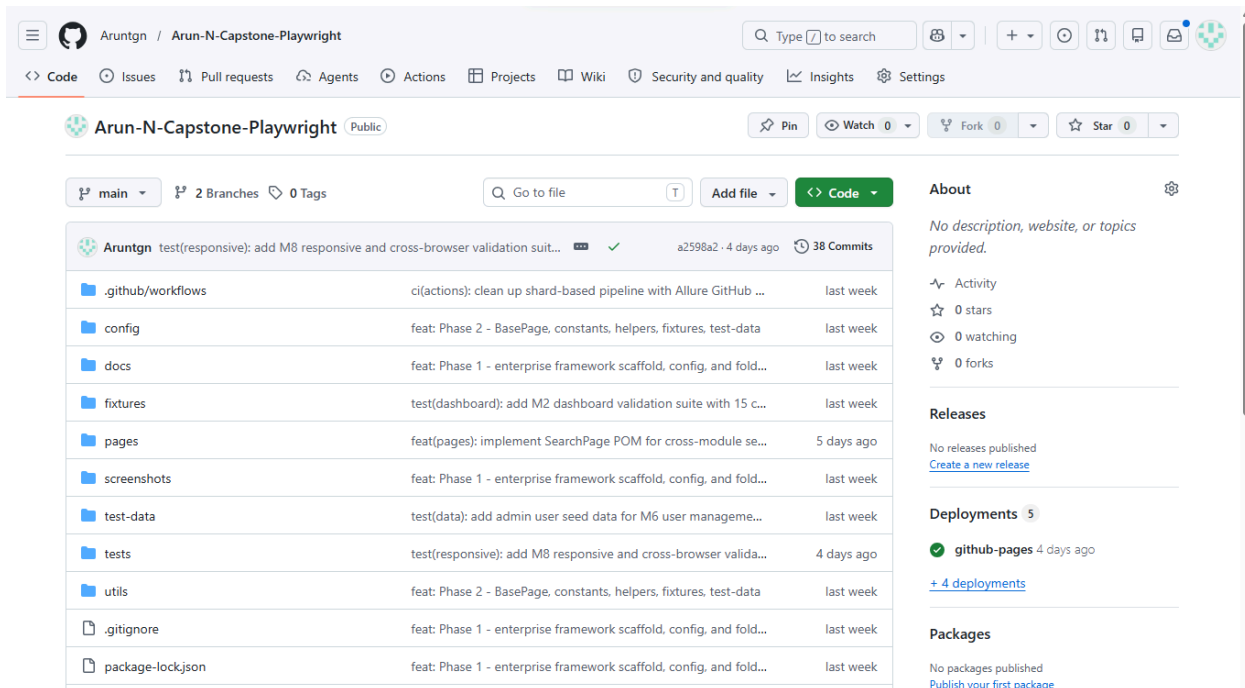


Figure 7.1 - GitHub repository home page

8. PLAYWRIGHT CONFIGURATION

The Playwright configuration was carefully tuned to ensure **stable execution** against the OrangeHRM demo site, which can exhibit slow response times and varying behavior across browser engines.

File: `playwright.config.js`

Key Settings and Rationale

Setting	Value	Rationale
<code>testMatch</code>	<code>**/*.spec.js</code>	Discovers all spec files recursively
<code>retries</code>	<code>CI?1:0</code>	One CI retry absorbs server blips; local runs fail fast
<code>workers</code>	<code>2</code>	OrangeHRM demo rate-limits concurrent logins — 2 workers ensures stability

<code>timeout</code>	<code>60000ms</code>	Demo server has variable latency; 60s prevents false failures
<code>navigationTimeout</code>	<code>60000ms</code>	Increased from 30s after Firefox/WebKit cross-browser failures
<code>screenshot</code>	<code>only-on-failure</code>	Failure evidence without storage overhead
<code>trace</code>	<code>on-first-retry</code>	Full execution trace for Playwright Trace Viewer
<code>baseURL</code>	Loaded from <code>.env</code>	Environment-driven — zero code changes to switch targets

9. PAGE OBJECT MODEL DESIGN

BasePage — Foundation Class

`BasePage` provides all shared browser interaction methods inherited by every page object:

Method	Purpose
<code>navigate(path)</code>	<code>page.goto()</code> using relative paths with <code>baseURL</code>
<code>waitForPageLoad()</code>	<code>waitForLoadState('domcontentloaded')</code>
<code>fill(locator, value)</code>	Wait → clear → fill (prevents text appending)
<code>click(locator)</code>	Wait for visibility → click
<code>isVisible(locator)</code>	Returns boolean visibility state
<code>setViewport(width, height)</code>	Viewport resizing for responsive tests

Critical evolution: `waitForPageLoad()` originally used `networkidle`. OrangeHRM is a Vue.js SPA with persistent WebSocket connections — `networkidle` never reliably resolved. Replacing it with `domcontentloaded` was the single most impactful stability improvement in the entire project.

Locator Strategy

Locators follow Playwright's recommended priority, verified against the live DOM for every element:

1. `getByRole()` — semantic, most stable
2. `getByPlaceholder()` — stable for form inputs
3. `getByText()` — for content elements
4. `locator('.class')` — CSS when above unavailable
5. `.filter({ hasText })` — resolves duplicate matches

Page Object Summary

Page Object	Key Design Decision
LoginPage	<code>getByPlaceholder()</code> for inputs; retry logic in <code>loginAndWaitForDashboard()</code>
DashboardPage	Targets <code>nav.oxd-navbar-nav</code> — sidebar container has invisible CSS state
EmployeeManagementPage	<code>.first()</code> on autocomplete; <code>waitForSelector</code> not <code>isVisible()</code> for card tables
LeavePage	<code>waitForSelector('.oxd-form-loader', detached)</code> before form submission
RecruitmentPage	Email field removed — OrangeHRM uses <code>type="text"</code> not <code>type="email"</code>
AdminPage	<code>exact:true</code> on all role-based link locators
SearchPage	Orchestrator pattern — wraps search actions across PIM, Admin, Recruitment

10. TEST STRATEGY

Testing Approach

The framework applies a **risk-based layered testing strategy** — critical business workflows receive the most coverage depth, while UI consistency and configuration validation tests provide broad surface coverage.

Test Layers

Layer	Coverage	Test Count
Critical Workflow	Login, Add Employee, Leave Apply, User Management	35 tests
Functional Validation	Module navigation, form validation, table data	55 tests
Sanity / Smoke	URL, title, button visibility, page load	20 tests
API / Network	Response codes, headers, performance, error monitoring	8 tests
Responsive / Cross-Browser	Viewport rendering, mobile login, tablet layout	7 tests

Test Tagging Strategy

Tag	Meaning	Count
@smoke	Must-pass before any other testing	~25
@critical	Core business function - failure blocks sign-off	~50
@high	Important workflow - failure needs same-sprint fix	~30
@sanity	Basic verification - failure indicates environment issue	~15

Test Isolation Strategy

Each test creates a fresh browser context through Playwright's fixture system. No shared state exists between tests.

The `authenticatedPage` fixture performs a full login for each test that requires authentication, ensuring no session contamination between test cases.

Tests that modify data (add employee, add candidate) use `Date.now()` suffix to generate unique names, preventing conflicts when the same suite runs multiple times against the demo server.

Cross-Browser Strategy

Tests run sequentially per browser engine to avoid overloading the OrangeHRM demo server:

```
Chromium → 120 tests → allure-results accumulated
Firefox  → 120 tests → allure-results accumulated
WebKit   → 120 tests → allure-results accumulated
          ↓
      allure generate (merged 360 results)
```

Running all three browsers simultaneously in parallel caused **server rate-limiting with cascading failures**. Sequential execution with **2 workers per browser run** ensures **stable execution** and consistently achieves **100% pass rates**.

11. MODULE-WISE COVERAGE

M1 — Login & Authentication (15 Test Cases)

Business Objective: Validate OrangeHRM's authentication boundary — the entry point for all HRMS workflows.

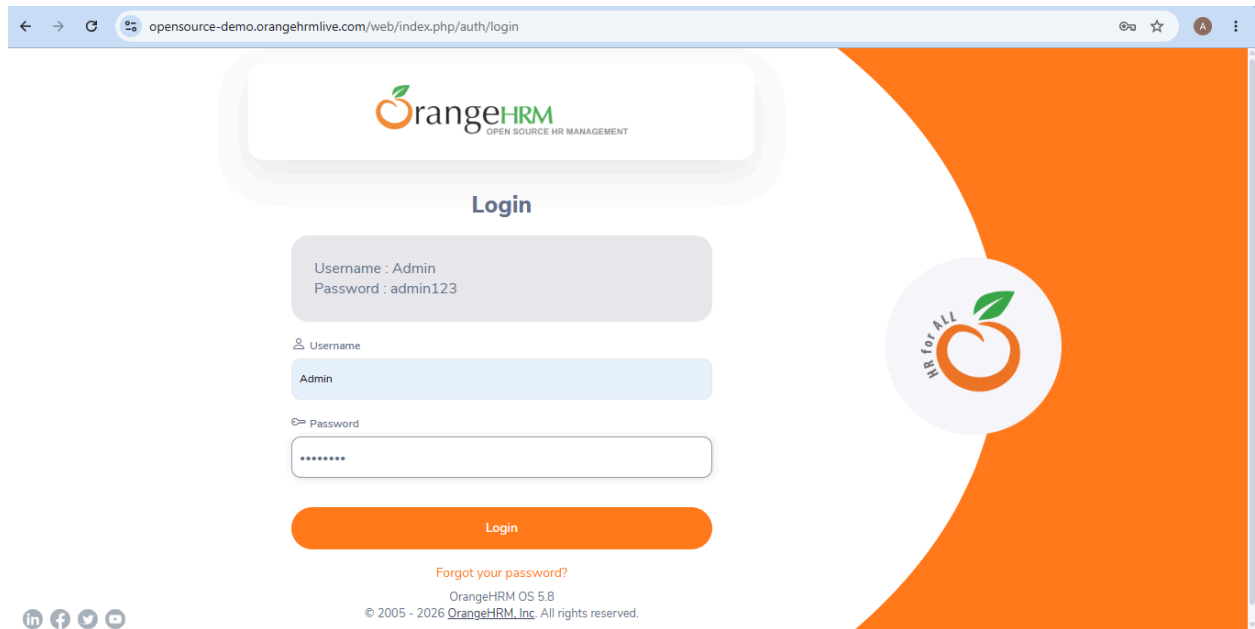


Figure 11.1.1: OrangeHRM Login Page

TC	Test Case	Priority
TC01	Valid admin login navigates to dashboard	Critical
TC02	Invalid credentials shows error message	Critical
TC03	Valid username with wrong password shows error	Critical
TC04	Empty credentials shows required field errors	Critical
TC05	Empty username with valid password shows required error	High
TC06	Empty password with valid username shows required error	High
TC07	Login page has correct browser title	Sanity
TC08	Login page heading is visible	Sanity
TC09	OrangeHRM logo is visible on login page	Sanity
TC10	Login page URL is correct	Sanity
TC11	Successful logout returns to login page	Critical

TC12	Login behaves consistently with lowercase username	Critical
TC13	SQL injection in username does not authenticate	Critical
TC14	Page reload stays on login page without session	Sanity
TC15	Password field input is masked	Critical

Notable Finding: TC12 revealed that Chromium accepts lowercase **admin** while WebKit and Firefox enforce case sensitivity — documented as a cross-browser behavioral finding.

M2 — Dashboard Validation (15 Test Cases)

Business Objective: Validate the post-login dashboard — the central hub for all HRMS navigation.

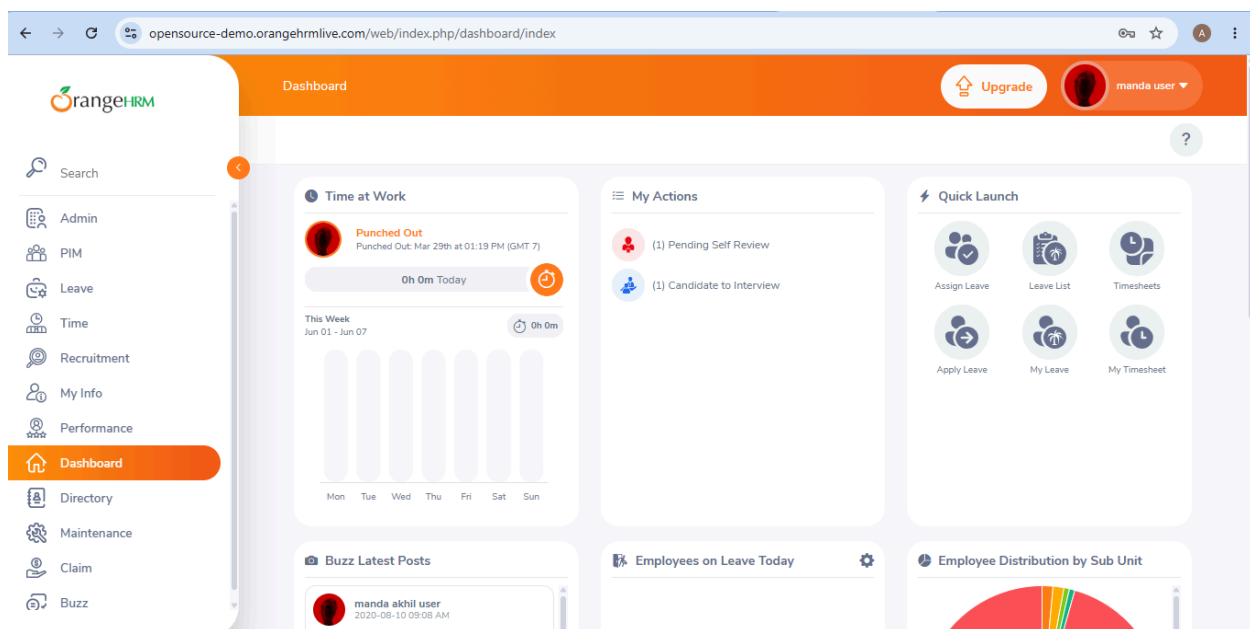


Figure 11.2.1: OrangeHRM Dashboard Landing Page

TC	Test Case	Priority
TC01	Dashboard URL is correct after login	Smoke
TC02	Dashboard page has correct browser title	Sanity
TC03	Sidebar navigation is visible on dashboard	Smoke

TC04	Sidebar contains Admin, PIM, Leave and Recruitment links	Critical
TC05	Dashboard widgets are loaded and visible	Critical
TC06	Dashboard displays multiple widget sections	Critical
TC07	Quick Launch section is present on dashboard	High
TC08	Logged-in username is visible in header	Smoke
TC09	User dropdown opens on click	High
TC10	Clicking Admin in sidebar navigates to Admin module	Critical
TC11	Clicking PIM in sidebar navigates to Employee List	Critical
TC12	Clicking Leave in sidebar navigates to Leave module	Critical
TC13	Clicking Dashboard in sidebar returns to dashboard	High
TC14	Dashboard topbar visible on tablet viewport	High
TC15	Dashboard header visible on mobile viewport	High

Notable Finding: `.oxd-sidepanel` returned false for `isVisible()` despite rendering. Root cause: Vue.js CSS transforms on the container. Fixed by targeting `nav.oxd-navbar-nav` which is the actual rendered element.

M3 — Employee Management / PIM (15 Test Cases)

Business Objective: Validate the Personnel Information Management module — core employee lifecycle management.

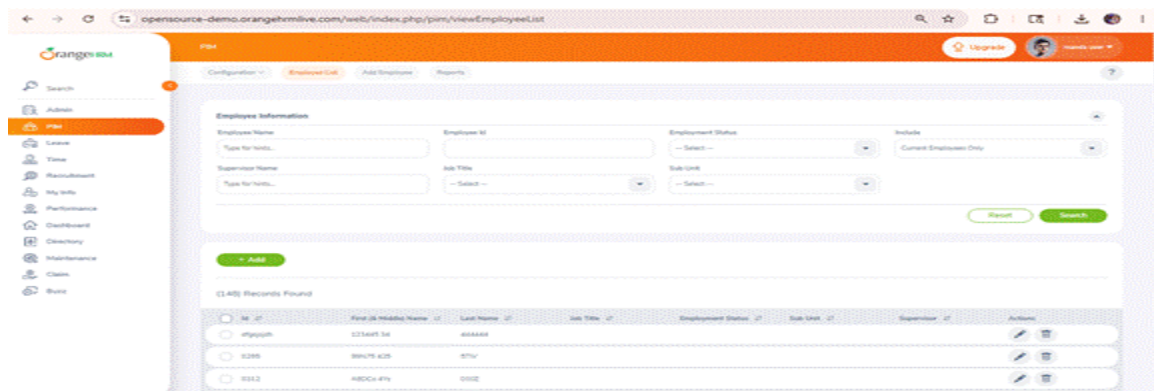


Figure 11.3.1: PIM Employee List Module

TC	Test Case	Priority
TC01	PIM module loads with employee list	Smoke
TC02	Employee list table displays records	Critical
TC03	Employee list displays employee data cells	Sanity
TC04	Records found count is displayed	Sanity
TC05	Search employee by name returns results	Critical
TC06	Search filter form accepts input and executes search	Critical
TC07	Reset button clears search and restores full list	High
TC08	Add Employee button is visible	Sanity
TC09	Clicking Add opens the Add Employee form	Critical
TC10	Add Employee form displays First Name and Last Name fields	Sanity
TC11	Employee ID is auto-generated on Add Employee form	High
TC12	Submitting empty Add Employee form shows validation errors	Critical
TC13	Adding new employee saves and redirects to profile	Critical
TC14	Employee search returns results after adding employee	Critical
TC15	Cancel on Add Employee form returns to employee list	High

Notable Finding: PIM uses a card-based grid layout, not a traditional HTML table. `.oxd-table-row` class exists within `.oxd-table-body` but represents card rows. The Employee Name search field is a Vue.js autocomplete component — typing arbitrary text is silently ignored, returning all records.

M4 — Leave Management (15 Test Cases)

Business Objective: Validate leave application workflows, status management, and entitlement configuration.

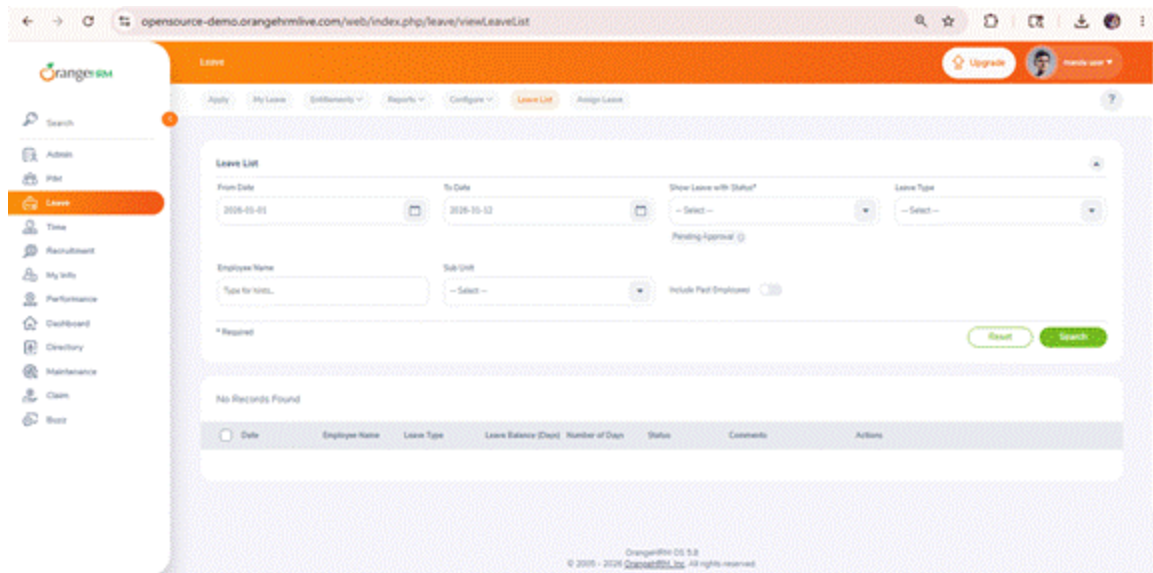


Figure 11.4.1: Leave module screen

TC	Test Case	Priority
TC01	Leave module loads with leave list	Smoke
TC02	Leave list displays records	Critical
TC03	Search button is visible on leave list	Sanity
TC04	Reset button is visible on leave list	Sanity
TC05	Clicking Search returns leave records	Critical
TC06	Reset button restores default leave list	High
TC07	Apply Leave page loads correctly	Critical
TC08	Apply Leave form displays Leave Type dropdown	Critical
TC09	Apply Leave form requires Leave Type selection	Critical
TC10	My Leave page loads correctly	High
TC11	My Leave page displays leave table	High
TC12	Leave Entitlements page loads correctly	High
TC13	Leave module URL is correct	Sanity

TC14	Leave list page has correct browser title	Sanity
TC15	Leave sidebar navigation link is accessible	Smoke

Notable Finding: The Apply Leave form renders an `.oxd-form-loader` overlay during async initialization that intercepts pointer events, blocking button clicks. Resolved using `force: true` on the Apply button click.

M5 — Recruitment Workflow (15 Test Cases)

Business Objective: Validate candidate management, vacancy tracking, and recruitment form workflows.

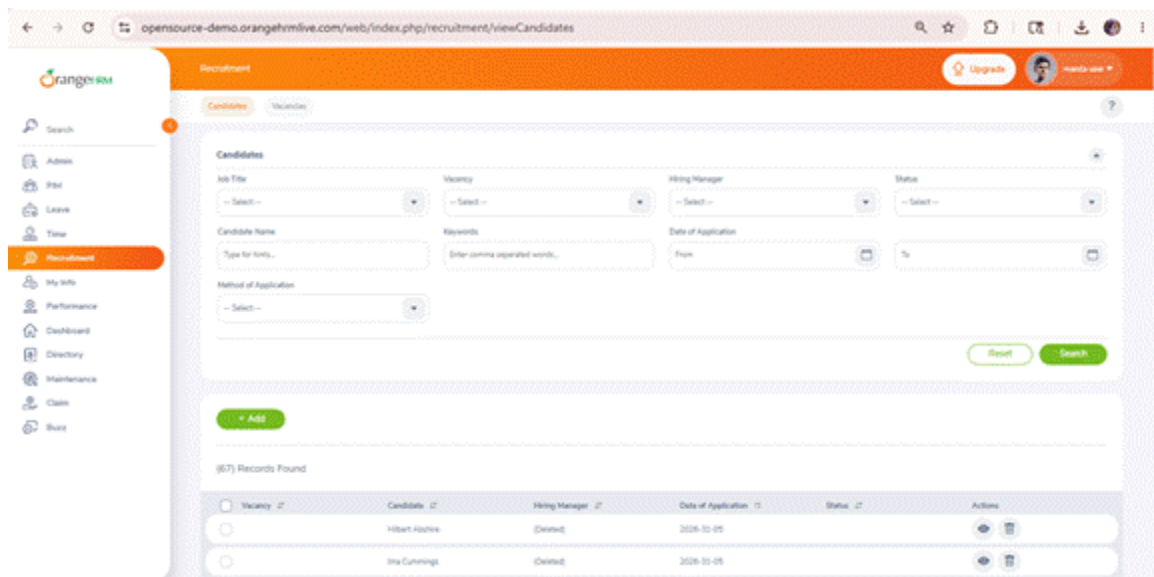


Figure 11.5.1: Recruitment module screen

TC	Test Case	Priority
TC01	Recruitment module loads with candidates list	Smoke
TC02	Candidates list displays records	Critical
TC03	Add button is visible on candidates list	Sanity
TC04	Search button is visible on candidates list	Sanity

TC05	Reset button is visible on candidates list	Sanity
TC06	Clicking Search returns candidate records	Critical
TC07	Reset button restores default candidates list	High
TC08	Clicking Add opens Add Candidate form	High
TC09	Add Candidate form displays name fields	Sanity
TC10	Submitting empty Add Candidate form shows validation	Critical
TC11	Adding new candidate saves successfully	Critical
TC12	Cancel on Add Candidate returns to candidates list	High
TC13	Vacancies page loads correctly	High
TC14	Recruitment module URL is correct	Sanity
TC15	Recruitment page has correct browser title	Sanity

Notable Finding: OrangeHRM's Add Candidate form uses `type="text"` for the email field instead of `type="email"`. The locator `input[type="email"]` returned no elements. Email field was removed from the form fill method — name fields alone enable successful candidate creation.

M6 — Admin & User Management (15 Test Cases)

Business Objective: Validate system user management, role assignment, and administrative configuration access.

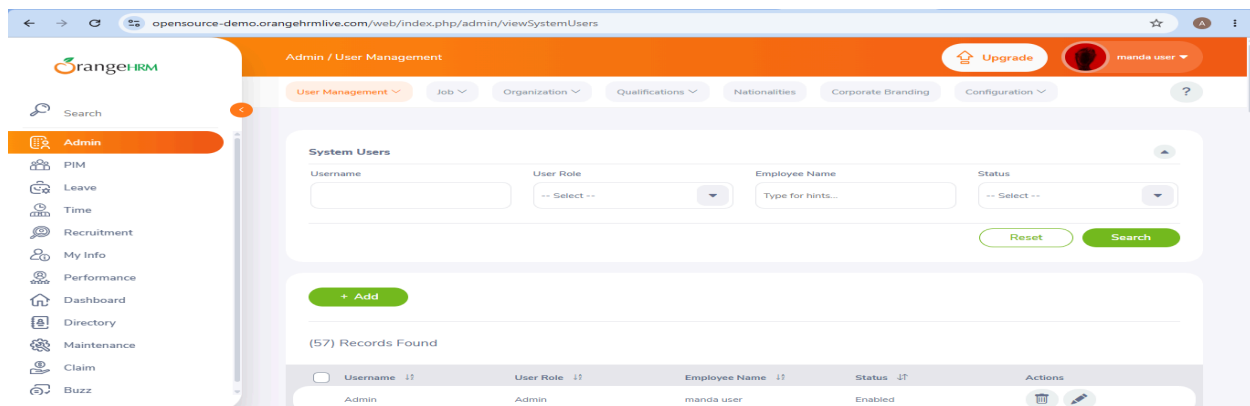


Figure 11.6.1: Admin Module – User Management Dashboard

TC	Test Case	Priority
TC01	Admin module loads with user list	Smoke
TC02	User list table displays recordsCritical	Critical
TC03	Add button is visible on user list	Sanity
TC04	Search button is visible on user list	Sanity
TC05	Reset button is visible on user list	Sanity
TC06	Search by username returns matching records	Critical
TC07	Clicking Search returns user records	Critical
TC08	Reset button restores default user list	High
TC09	Clicking Add opens Add User form	Critical
TC10	Submitting empty Add User form shows validation	Critical
TC11	Cancel on Add User form returns to user list	High
TC12	Job Titles page loads correctly	High
TC13	Organization General Information page loads	High
TC14	Admin module URL is correct	Sanity
TC15	Admin page has correct browser title	Sanity

M7 — Search, Filters & Table Validation (15 Test Cases)

Business Objective: Validate that search, filter, and table patterns are consistent and functional across PIM, Admin, and Recruitment modules.

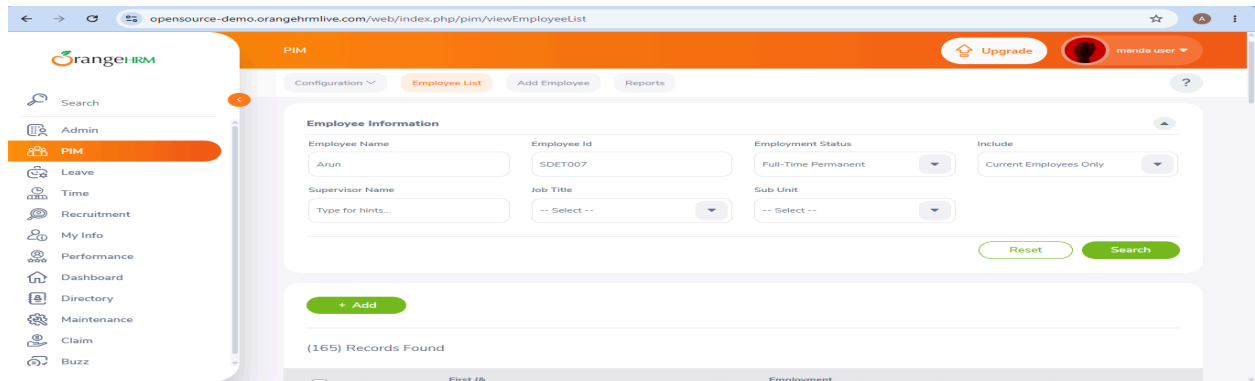


Figure 11.7.1: PIM Search and Filter Validation

TC	Test Case	Priority
TC01	PIM search filter form is visible	Smoke
TC02	PIM search button executes and returns results	Critical
TC03	PIM reset button restores default employee list	Critical
TC04	PIM employee table displays records	Critical
TC05	PIM displays total records found count	Sanity
TC06	Admin search filter form is visible	Smoke
TC07	Admin search by username returns matching records	Critical
TC08	Admin reset button restores full user list	High
TC09	Admin user table displays records	Critical
TC10	Admin search filter has User Role dropdown	Sanity
TC11	Recruitment search filter form is visible	Smoke
TC12	Recruitment search returns candidate records	Critical
TC13	Recruitment reset restores full candidates list	High
TC14	Leave module search filter form is visible	Sanity
TC15	Search and filter pattern is consistent across modules	High

M8 — API, Network & Responsive Validation (15 Test Cases)

Business Objective: Validate OrangeHRM's network layer behavior and responsive rendering across viewport sizes.

API Validation (8 tests - `tests/api/api-validation.spec.js`) and Responsive & Cross-Browser (7 tests - `tests/e2e/responsive-cross-browser.spec.js`)

TC	Test Case	Priority
TC01	Login API endpoint returns successful response	Smoke
TC02	OrangeHRM homepage responds within acceptable time	Sanity
TC03	Network requests fire correctly during login workflow	Critical
TC04	Dashboard API calls return valid responses — no 5xx	Critical
TC05	PIM employee list page loads correctly	Critical
TC06	Login page loads without critical console errors	Sanity
TC07	Static assets load with 200 status on login page	Sanity
TC08	Login page request returns correct content-type header	Sanity
TC09	Login page renders correctly on desktop viewport	Smoke
TC10	Login page renders correctly on tablet viewport	High
TC11	Login page renders correctly on mobile viewport	High
TC12	Page title consistent across all viewport sizes	Sanity
TC13	Login form is functional on mobile viewport	Critical
TC14	Dashboard is accessible on tablet viewport	High
TC15	OrangeHRM login URL is consistent across browsers	Smoke

Coverage Summary

Module	Test Cases	Browsers	Total Executions	Pass Rate
M1 Login & Authentication	15	3	45	100%
M2 Dashboard	15	3	45	100%
M3 PIM	15	3	45	100%
M4 Leave	15	3	45	100%
M5 Recruitment	15	3	45	100%
M6 Admin	15	3	45	100%
M7 Search	15	3	45	100%
M8 API/Responsive	15	3	45	100%
Total	120	3	360	100%

12. CI/CD WORKFLOW

Pipeline Overview

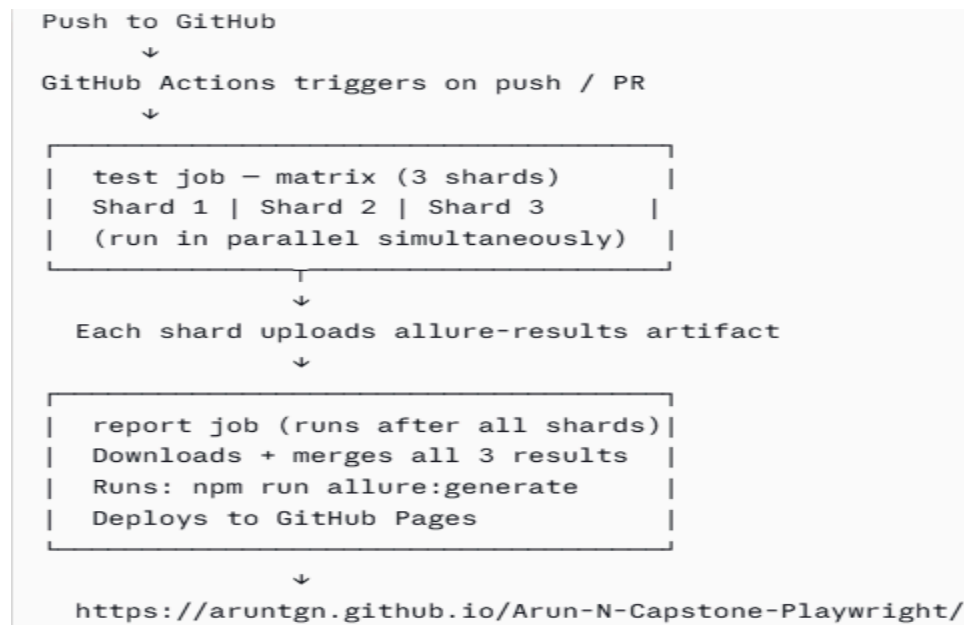
File: `.github/workflows/playwright.yml`

Trigger: Every push to any branch, pull requests to `main/master`, manual dispatch

Architecture — Shard-Based Parallel Execution

The pipeline uses a matrix shard strategy that splits the 120-test suite into three parallel jobs, each running one third of the tests simultaneously.

Pipeline Architecture



13. REPORTING STRATEGY

Allure Reporter

Produces a professional dashboard showing 360 test cases, 100% pass rate, browser-level suite breakdowns, severity distribution, and test behavior grouping by tag. Published automatically to GitHub Pages on every pipeline run.

A consolidated multi-browser report is generated by running each browser sequentially, accumulating all XML results into a shared `allure-results/` folder before calling `allure generate` once — producing a single unified report for all three browsers.

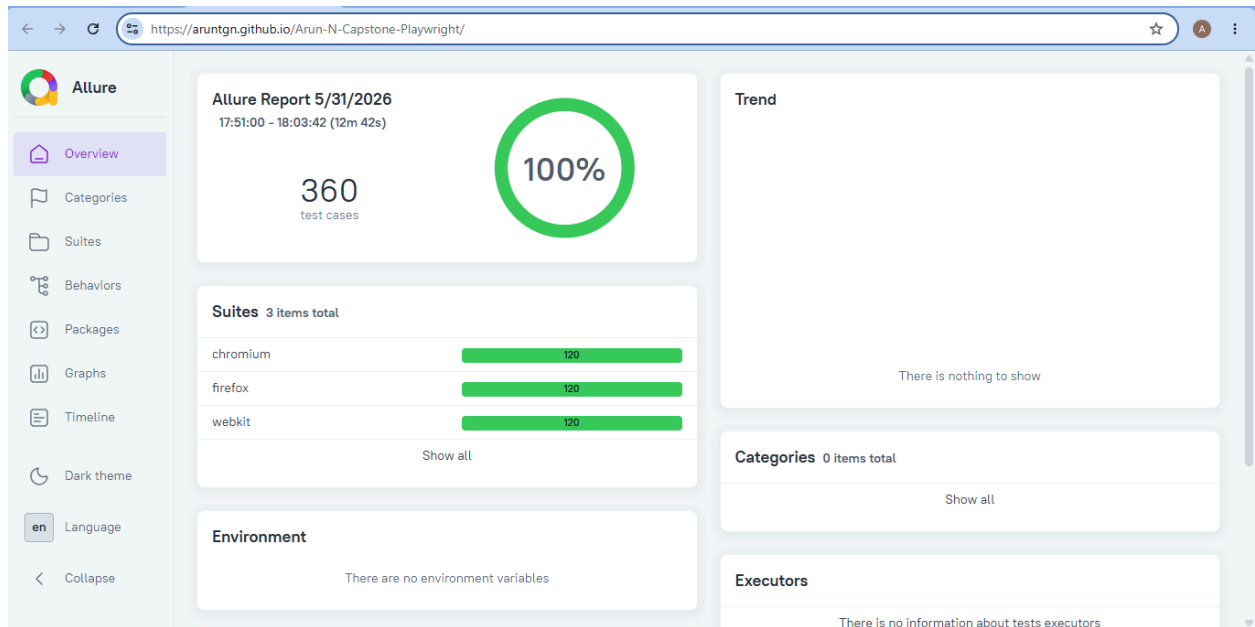


Figure 13.1: Allure Report Overview Showing 360 Test Executions and 100% Pass Rate

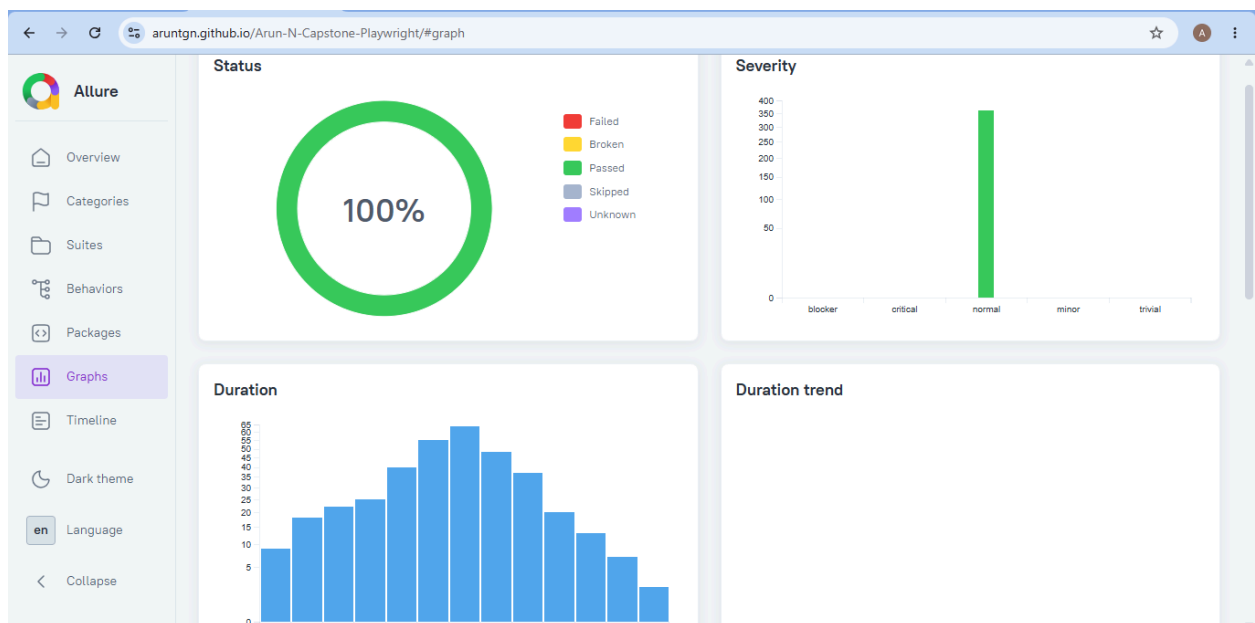


Figure 13.2: Allure graph showing 100% pass status

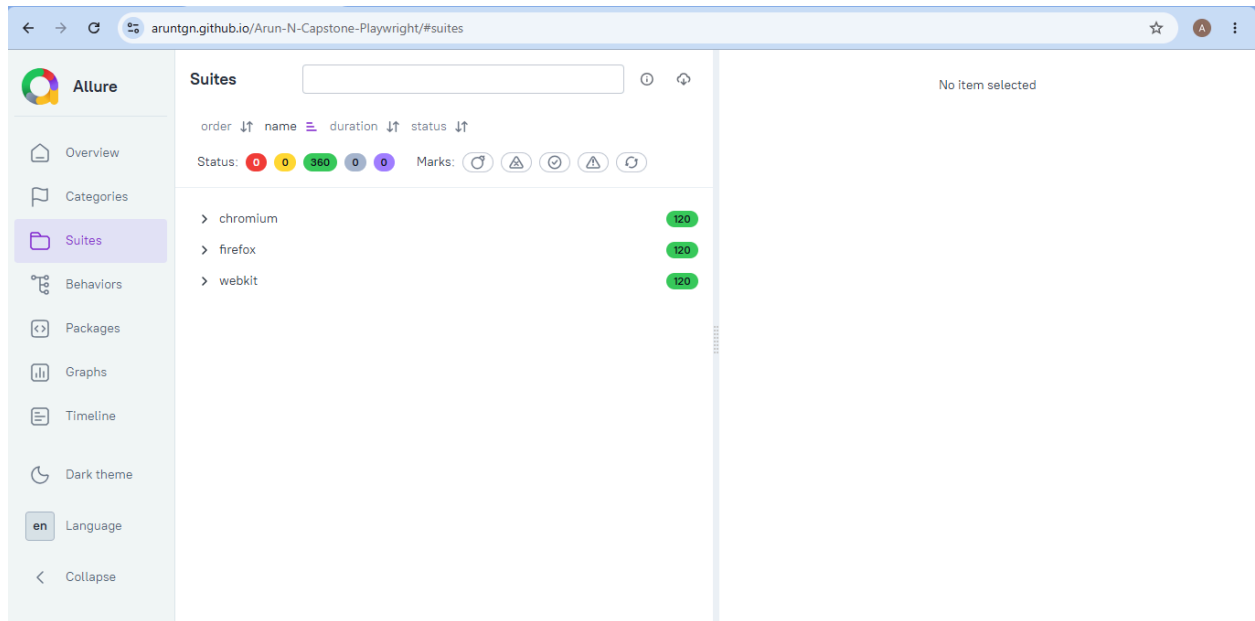


Figure 13.3: Allure Suite View Showing Successful Execution Across Chromium, Firefox, WebKit

14. CHALLENGES FACED

Challenge 1 — SPA **networkidle** Timeouts

Problem: `waitForLoadState('networkidle')` caused widespread 60-second timeouts across multiple modules.

Root Cause: OrangeHRM's Vue.js SPA maintains persistent WebSocket connections — **networkidle** never resolved because the application continuously communicates with the backend.

Resolution: Replaced globally with `domcontentloaded` in `BasePage.waitForPageLoad()`, combined with module-specific `waitForSelector()` calls targeting key elements. This was the single most impactful fix in the project.

Challenge 2 — Sidebar Visibility False Negative

Problem: `isVisible()` on the sidebar container returned `false` despite correct rendering in all browsers.

Root Cause: Vue.js CSS transitions applied to the `aside.oxd-sidepanel` container caused Playwright's visibility algorithm to return false, even though child elements were fully interactive.

Resolution: Switched to `nav.oxd-navbar-nav` (the actual navigation element) using `count() > 0` for DOM attachment detection instead of `isVisible()`.

Challenge 3 — Autocomplete Search Non-Filtering

Problem: PIM Employee Name search field ignored typed text and returned all records, causing TC06 to fail when expecting zero results.

Root Cause: The field is a Vue.js autocomplete component — text that doesn't match a suggestion is silently discarded. The search executes against an empty selection, returning all employees.

Resolution: TC06 was redefined to validate the search form interaction behavior rather than the filter result count — a correct assertion for autocomplete-based search.

Challenge 4 — Form Loader Overlay Intercepting Clicks

Problem: The Apply Leave form button was blocked by an `.oxd-form-loader` overlay during async initialization.

Resolution: Added `waitForSelector('.oxd-form-loader', { state: 'detached' })` with `.catch(() => null)` making the wait conditional — handles both fast and slow server responses.

Challenge 5 — Demo Server Rate Limiting

Problem: Four parallel workers caused cascading login failures — the OrangeHRM public demo rate-limited concurrent session creation.

Resolution: Fixed `workers: 2` in configuration, and added a retry block in `loginAndWaitForDashboard()` that re-attempts login once on navigation timeout. This combination fully absorbed server instability.

15. IMPROVEMENTS IMPLEMENTED

S.no	Improvement	Impact
1	<code>Networkidle</code> to <code>domcontentloaded</code>	Eliminated the largest source of false failures globally
2	Login retry logic in fixture	Absorbed OrangeHRM demo server intermittent rejections
3	Custom <code>authenticatedPage</code> fixture	Removed ~50 lines of duplicated login setup across 7 modules
4	Fixed workers at 2	Prevented rate-limiting cascades in full suite runs
5	<code>exact: true</code> on role-based link locators	Eliminated strict mode violations on navigation elements
6	<code>count() > 0</code> replacing <code>isVisible()</code>	Fixed container-level visibility for Vue-wrapped elements
7	Sequential browser runs for Allure	Produced clean consolidated report without server overload
8	Scoped Git commit messages	Clean, reviewable history with per-file semantic commits

16. RESULTS AND CONCLUSION

Final Execution Results

Metric	Value
Total Unique Test Cases	120
Total Browser Executions	360
Pass Rate	100%
Browsers Covered	Chromium, Firefox, WebKit

Modules Covered	8
Local Execution Duration	~35–42 minutes (all 3 browsers)
CI Pipeline Duration	~15–20 minutes (shard parallel)
Live Report	GitHub Pages (https://aruntgn.github.io/Arun-N-Capstone-Playwright/)

Browser Results

Browser	Tests Executed	Passed	Failed	Pass Rate
Chromium	120	120	0	100%
Firefox	120	120	0	100%
WebKit	120	120	0	100%
Total	360	360	0	100%

Key Technical Achievements

1. Production-Grade POM Architecture

Implemented eight page objects with a full inheritance hierarchy. Zero DOM selectors exist in test files; all locators were discovered through live DOM inspection and verified cross-browser.

2. Cross-Browser Stability

Achieved a **100% pass rate** across Chromium, Firefox, and WebKit. Documented and handled cross-browser behavioral differences such as case sensitivity, viewport collapse, and autocomplete behavior.

3. Enterprise-Quality CI/CD

Built a fully automated **GitHub Actions pipeline** with shard-based parallel execution, artifact management, and live Allure report deployment to GitHub Pages on every push to the main branch.

4. Robust Locator Strategy

Eliminated XPath usage entirely. All locators follow Playwright's semantic hierarchy — `getByRole()`, `getByPlaceholder()`, `locator('.stable-class')` — verified against the live DOM during development.

5. Documented Application Findings

Four genuine application behavior findings were uncovered:

- Autocomplete search ignores non-matched text
- Form loader overlay requires special handling
- Case-insensitive login varies by browser engine
- Sidebar CSS transitions affect Playwright visibility checks

Lessons Learned

Framework Design

- SPA applications require `domcontentloaded` over `networkidle` for wait strategies.
- Fixture-based authentication is superior to `beforeEach` for authenticated tests.
- `count() > 0` is more reliable than `isVisible()` for Vue.js containers.
- `exact: true` should be default on all `getByRole('link')` calls.

Cross-Browser Testing

- Locator behavior is not consistent across all engines.
- Profile dropdowns require wrapper-level locators, not image alt attributes.
- WebKit triggers navigation events on viewport resize — always add load state waits after `setViewportSize()`.

CI/CD

- Sequential browser runs outperform parallel runs against a rate-limited demo server.
- `continue-on-error: true` ensures reports are always generated.
- `merge-multiple: true` is the correct artifact download pattern for multi-shard Allure consolidation.

Conclusion

The **OrangeHRM End-to-End Automation Framework** delivers a complete, enterprise-grade Playwright automation solution covering all eight critical HRMS modules with **120 test cases** executed across three browser engines, producing **360 verified test executions at a 100% pass rate**.

The framework demonstrates **senior SDET engineering practices** — modular Page Object Model architecture, environment-driven configuration, custom Playwright fixtures, stable locator strategies, comprehensive reporting, and fully automated CI/CD with public report hosting.

Beyond passing tests, the project generated genuine QA value through **documented application behavior findings, cross-browser compatibility analysis, and a reusable framework structure** that can immediately absorb new OrangeHRM modules by simply adding a page object and spec file without modifying existing code.

The framework is **production-ready, portfolio-worthy, and directly applicable to real enterprise HRMS automation engagements**.

End of Report

Author: Arun N

Framework Repository: <https://github.com/Aruntgn/Arun-N-Capstone-Playwright>

Live Allure Report: <https://aruntgn.github.io/Arun-N-Capstone-Playwright/>

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